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OpenBalance: An Interoperable Computational Framework, Progressive 8-Stage Protocol, and Outlier Metric for Dynamic Posturography—A Technical Proof-of-Concept

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Highlights

What are the main findings?

- The open-source, hardware-interoperable OpenBalance framework standardizes posturographic analysis, illustrating how the choice of mathematical model leads to varying 'Sway Area' results for the exact same dataset.
- A dynamic 8-stage protocol and the Outlier Excursion Ratio (OER) are designed to quantify corrective movements and potentially reduce ceiling effects in cohorts with high baseline balance ability.

What are the implications of the main findings?

- By processing raw data from different force plates via a unified Python library, the framework provides an accessible open-source processing pipeline to facilitate cross-study comparability.
- The progressive protocol and OER provide a standardized framework intended to increase postural challenge beyond static baseline conditions.

Abstract

Human postural control assessment is currently characterized by methodological inconsistencies in data processing, measurement protocols, and spatial sway area calculations, often resulting in potential ceiling effects when evaluating cohorts with high baseline balance stability. To address these limitations, we developed OpenBalance, an interoperable, open-source Python framework designed to standardize signal processing across diverse commercial and custom force plate formats. Alongside this computational architecture, we propose a progressive 8-stage measurement protocol designed to scale postural difficulty and introduce a dimensionless metric, the Outlier Excursion Ratio (OER). The OER mathematically leverages the spatial disparity between the Convex Hull Area and the 95% Prediction Ellipse Area to quantify extreme spatial trajectory excursions. A single-subject dataset was processed as a technical proof-of-concept to verify the software pipeline's execution and parameter calculation under empirical measurement conditions. Furthermore, automated extraction demonstrated the OER to be mathematically sensitive to transient spatial trajectory excursions, illustrating the methodological inconsistencies of comparing ambiguous sway area metrics across studies. By shifting from rigid data-upload services to an adaptable modular library, the OpenBalance framework provides an open-access infrastructure to support standardized signal processing and reproducible data analysis.

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1. Introduction

Human postural control is a fundamental motor skill essential for navigating daily activities, preventing falls in clinical populations, and maximizing performance in elite sports. The objective assessment of human balance relies on a variety of methods. In computerized posturography, ground reaction forces are frequently measured to evaluate postural control. Force plate systems are predominantly used for this purpose due to their high spatial and temporal resolution [1–3]. However, beyond the technical hardware requirements, study design and data analysis play a decisive role. As demonstrated by comprehensive systematic reviews, the reliability of posturographic parameters depends substantially on measurement duration and the number of trials, with experts recommending multiple measurement repetitions to achieve acceptable test–retest reliability [4]. Commercial systems have established extensive, standardized batteries to address this [5–8].

Standardization across measurement protocols remains a critical challenge in posturography. Regarding study design, measurement protocols are often strictly tailored to the physical limitations of specific patient groups. While this is ethically and practically understandable, such designs lose their comparability when evaluated against healthy control groups. Specifically, optimizing protocols for pathological conditions reduces the physical demand of the measurement. Consequently, studies risk a ceiling effect: post-intervention patients might appear to have reached the level of a healthy cohort simply because the test lacks the difficulty to reveal further differences. To differentiate between healthy individuals—or to compare healthy subjects with elite athletes—a substantially dynamic and challenging measurement setup is required. This implies that longitudinal studies may need adaptive or progressive measurement protocols, adjusting the difficulty from baseline to post-intervention, to capture actual physiological changes and mitigate potential ceiling effects [9–11].

Furthermore, inconsistencies in data processing hinder study comparability. Different research groups employ varying preprocessing techniques, such as signal filtering and scaling. Moreover, disparate mathematical methodologies are applied to calculate identical metrics. A prominent example is the Sway Area (often referred to as Range of Motion), which is inconsistently calculated using confidence circles, confidence ellipses, or full convex hulls, leading to quantitatively distinct results for identical datasets [12–15]. However, while these divergent models are typically viewed as a source of methodological inconsistency in the literature, their mathematical disparity actually holds untapped biomechanical value: the discrepancy between statistically filtered core sway (Ellipse) and absolute trajectory bounds (Hull) can be utilized to quantify the magnitude of sporadic corrective movements.

To standardize parameter evaluation and address the aforementioned methodological limitations, this paper introduces a modular, open-source software framework. Specifically, the contributions of this study are threefold:

1. We present an interoperable software architecture (OpenBalance) that standardizes signal processing and parameter extraction across established commercial force plate data formats.
2. We propose a progressive, dynamic 8-stage measurement protocol designed to systematically scale postural challenge, which is intended to extend the dynamic range and potentially mitigate ceiling effects in balance assessment.

3. We introduce a dimensionless metric—the Outlier Excursion Ratio (OER)—which mathematically leverages the spatial disparity between different Sway Area models to quantify extreme spatial trajectory excursions.

To foster transparency and community-driven development, the complete computational framework and empirical datasets are provided open-access.

2. Materials and Methods

2.1. Software Architecture, Modularity, and Distribution

To address methodological heterogeneity in posturographic data processing, we developed OpenBalance, a scalable, object-oriented software framework written in Python version 3.14.5 (Python Software Foundation, Wilmington, DE, USA). Python was selected as the foundational language due to its widespread adoption and extensive ecosystem in scientific computing and reproducible research [16].

2.1.1. Software Interoperability and Modular Input Interfaces

The primary design paradigm of the framework adheres to core software engineering best practices, specifically the Dependency Inversion Principle (part of the SOLID principles) [17]. Instead of developing standalone, tightly coupled analysis scripts for specific force plate models, the architecture separates the data pipeline into distinct, independent modules: Data Input (Readers), Signal Processing (Filters and Kinematics), and Data Output (Visualization and Reporting).

The input layer is designed to dynamically import raw force and center of pressure (COP) data from four commercial systems (Kistler (Kistler, Winterthur, Switzerland), Tetrax (Sunlight Medical Ltd., Ramat Gan, Israel), Tymo (Tyromotion GmbH, Graz, Austria), and AMTI (AMTI, Watertown, MA, USA)), as well as custom-built hardware like the 16-cell OpenBalance platform [18]. By utilizing Python's Abstract Base Classes (ABCs), the system implements polymorphism to standardize incoming data into a uniform "Trial" object. This abstraction decouples downstream mathematical operations from proprietary file structures, hardware-specific spatial layouts, and sampling rates. Consequently, integrating support for an additional force plate system only requires the implementation of a single reader class without altering the underlying kinematic algorithms.

2.1.2. Output, Reporting, and Export

For clinical and research applications, the framework features an automated reporting pipeline. Results are exported as structured, machine-readable data for statistical analysis (e.g., CSV, JSON) and as comprehensive, multi-lingual PDF reports. To ensure visual comparability across different measurement conditions, these reports dynamically overlay different spatial models (e.g., Convex Hull, Prediction Ellipse) utilizing identical, globally calculated axis limits.

2.1.3. Deployment and Scientific Reproducibility

A recognized challenge in biomechanical computational research is the reported reproducibility challenges caused by missing dependencies or platform-specific execution environments [19]. To mitigate this, the OpenBalance architecture was converted from a local script repository to a standardized Python package. The software is published and persistently maintained on the Python Package Index (PyPI). This allows researchers to integrate the framework into their own environments via standard package managers (i.e., pip install openbalance), ensuring that all required dependencies (e.g., SciPy, NumPy) are automatically resolved. This centralized distribution model supports scientific reproducibility and establishes an accessible platform for future community-driven development.

2.2. Signal Processing and Posturographic Parameters

All kinematic parameters are derived through a standardized, cascading calculation pipeline to ensure reproducibility.

2.2.1. Preprocessing

Raw ground reaction force data is filtered using a zero-phase, second-order Butterworth low-pass filter with a configurable cutoff frequency (default: 13 Hz), following established biomechanical standards [2,20,21], to attenuate high-frequency noise and sensor artifacts. Because reactive equilibrium corrections may contain signal content up to 15 Hz [5,20,22,23], and because low-pass filtering attenuates the extreme excursions targeted by the OER (Section 2.2.4), the cutoff frequency is user-configurable and must be reported alongside any OER value. For multi-cell systems, the COP is subsequently calculated using the force-weighted average of the sensor coordinates.

2.2.2. Standard Kinematic Parameters

From the filtered COP trajectories [20], the system calculates established stability metrics, including the total COP Path Length and the Weight Distribution Index (WDI) [5]. The WDI quantifies the symmetry of weightbearing across four quadrants (left/right, anterior/posterior) and is expressed as a percentage deviation from an ideal 25% distribution per quadrant. The COP Path Length (PL) is a widely reported metric for assessing general postural activity (shown in Equation (1)). It represents the total distance traveled by the Center of Pressure over the duration of the measurement. PL is calculated as the sum of the Euclidean distances between consecutive COP coordinates:

$$\mathrm{PL} = \sum_{\mathrm{i}=1}^{\mathrm{N}-1} \sqrt{(x_{\mathrm{i}+1} - x_{\mathrm{i}})^2 + (y_{\mathrm{i}+1} - y_{\mathrm{i}})^2} \quad (1)$$

where N is the total number of recorded data points, and x_i and y_i represent the filtered COP coordinates at the i-th sample.

2.2.3. Advanced Sway Area Analysis

To address the methodological inconsistencies in spatial sway evaluation, the framework simultaneously calculates the Sway Area using three distinct mathematical models:

- 95% Confidence Circle: Based on the mean distance (MDIST) and the standard deviation of distances, as defined by Prieto et al. [13].
- 95% Prediction Ellipse: Utilizing the eigenvalues of the COP covariance matrix to capture the primary direction of sway, following Schubert and Kirchner [12].
- Convex Hull Area: Enclosing all measured COP points to capture the absolute maximum excursions [15].

2.2.4. Outlier Excursion Ratio (OER)

To quantify the relative spatial magnitude of transient trajectory excursions—which are often masked by the 95% Prediction Ellipse but captured by the Convex Hull—we introduce the Outlier Excursion Ratio. The OER is defined as the quotient of the Convex Hull Area and the 95% Prediction Ellipse Area. While an OER value near 1.0 indicates a consistent, stationary sway pattern, an elevated OER serves as a dimensionless metric for the spatial expansion of trajectory outliers. Traditional sway area models, such as the 95% Prediction Ellipse Area (PEA), effectively describe the core postural sway by statistically filtering out extreme values based on data covariance. Conversely, the Convex Hull Area (CHA) encloses the absolute spatial boundary of the COP trajectory, making it sensitive to singular, extreme data points.

To mathematically isolate and quantify these spatial trajectory excursions beyond the core sway area, we introduce two parameters derived from the juxtaposition of these geometric models.

The Outlier Excursion Area (OEA) represents the differential spatial area between the absolute boundary model and the statistical core distribution. It is calculated as the direct difference between the Convex Hull Area and the 95% Prediction Ellipse Area:

$$\mathrm{OEA} = \mathrm{CHA} - \mathrm{PEA}_{\{95\%\}} \tag{2}$$

A positive OEA (OEA > 0 cm^2, corresponding to CHA > PEA) quantifies the net spatial area spanned by extreme trajectory points extending beyond the 95% confidence ellipse. Conversely, when CHA < PEA, the OEA becomes negative (OEA < 0 cm^2), mathematically reflecting a tightly clustered trajectory that remains entirely enclosed within the statistical ellipse model without spatial outliers.

To minimize the influence of a subject's absolute sway magnitude (e.g., due to body height or general baseline sway) and provide a standardized, dimensionless metric, we define the OER as the quotient of the two models:

$$\mathrm{OER} = \frac{\mathrm{CHA}}{\mathrm{PEA}_{95\%}} \tag{3}$$

The OER serves as a normalized metric for relative spatial trajectory expansion. A ratio near 1.0 indicates a steady, uniformly distributed sway pattern where the Convex Hull closely approximates the Prediction Ellipse. In contrast, an elevated ratio (OER >> 1.0) indicates that the postural trajectory contained transient excursions of large spatial magnitude relative to the core sway ellipse.

Two mathematical properties of the OER must be considered when interpreting or comparing values. First, because the Convex Hull Area expands monotonically with additional samples while the Prediction Ellipse Area approaches an asymptotic threshold, the OER is inherently dependent on recording duration and sampling frequency; OER values are therefore only comparable across trials of identical, standardized duration (30 s in this work). Second, since the Prediction Ellipse is calculated via the covariance matrix and is not strictly bounded by observed sample points, OER values below 1.0 (and correspondingly negative OEA values) are mathematically possible for tightly clustered, stationary trajectories. In such cases (CHA < PEA), the observed trajectory falls well within the predicted statistical area, confirming the absence of spatial trajectory outliers.

2.3. Assessment Protocols and Study Design

The selection of an appropriate measurement protocol is critical to avoid ceiling effects, particularly when assessing healthy individuals or athletes. The complexity of posturographic assessments generally dictates the clinical relevance of the data.

2.3.1. Evolution of Static Measurement Protocols

The standard baseline assessment consists of a single static measurement under unconstrained bipedal stance. This is typically extended by altering visual input, resulting in consecutive measurements with eyes open (EO) and eyes closed (EC), commonly known as the Romberg test [22]. To further challenge the somatosensory system, researchers often repeat the EO and EC conditions on compliant foam pads, which degrade proprioceptive feedback from the plantar surface.

Commercial systems have established more extensive, standardized batteries. For instance, the Sensory Organization Test (SOT) by SmartEquitest (NeuroCom International, Inc., Clackamas, OR, USA) dynamically alters the visual surround and support surface [6],

while the static Tetrax protocol utilizes a sequence of eight specific measurements incorporating head rotations and foam surfaces to isolate vestibular, visual, and somatosensory deficits [5].

2.3.2. Progressive 8-Stage Protocol

However, conventional static protocols may not produce measurable differences when evaluating athletic or healthy cohorts, as the physiological challenge is insufficient to exhaust their compensatory reserves [10]. To address this, we propose an incrementally challenging 8-stage measurement protocol designed to systematically scale postural difficulty, thereby potentially mitigating ceiling effects in highly trained individuals [11], while remaining adaptable for research and clinical applications:

- Measurement 1 (M1): Bipedal stance, stable surface, Eyes Open (Baseline).
- Measurement 2 (M2): Bipedal stance, stable surface, Eyes Closed.
- Measurement 3 (M3): Bipedal stance on compliant foam pads (e.g., kybun (kybun Joya AG, Roggwil, Switzerland) mats), Eyes Open.
- Measurement 4 (M4): Bipedal stance on compliant foam pads, Eyes Closed.
- Measurements 5 to 8 (M5–M8): Repetition of conditions M1 through M4, but performed on a hemispherical, unstable base (wobble board/Kalotte).

To create an elevated postural challenge intended to expand the measurable dynamic range, measurements 5 through 8 transition from a static to an unstable support surface. To ensure sensor baseline stability, static zero-load offset measurements are performed immediately before and after the complete eight-stage series, and the resulting arithmetic mean is subtracted element-wise from all force channels. Given the increased postural instability during eyes-closed conditions on the unstable base (M6 and M8), all trials should be conducted with a spotter positioned within arm's reach and a stable support rail available to the subject; a trial is aborted and repeated if the subject requires external support. The complete matrix of the experimental conditions is detailed in Appendix A, and a visual quick-reference guide for laboratory execution is provided in Appendix B.

This progressive protocol is designed to allow baseline measurements (M1–M4) to be conducted under standard static conditions, while the advanced dynamic stages (M5–M8) introduce a multi-axial instability paradigm intended to evaluate dynamic postural control across a wider operational spectrum.

3. Results

3.1. Proof of Concept: The Progressive 8-Stage Protocol

To evaluate the technical feasibility of the proposed OpenBalance framework and the progressive 8-stage measurement protocol, the processing pipeline was applied to empirical posturographic data. Specifically, a proof-of-concept dataset was acquired from a single healthy adult volunteer to verify execution and data extraction across all eight protocol stages. Hardware setup, sensor calibration, and data acquisition procedures strictly followed the methodology detailed in our previous work [18].

As hypothesized, standard static measurements (M1: Stable Surface, Eyes Open) produced baseline postural sway, resulting in a tightly clustered Center of Pressure (COP) trajectory with a path length of 26.41 cm and a 95% Prediction Ellipse area of 1.36 cm^2. Introducing somatosensory and visual constraints on a stable surface (M4: Compliant Foam Pad, Eyes Closed) produced an intermediate increase in postural sway, yielding a path length of 149.95 cm and a prediction ellipse area of 23.32 cm^2. Conversely, advancing the protocol to the dynamic stages on the unstable hemispherical base (M5–M8) revealed a substantial increase in postural challenge and COP trajectory displacement. For instance,

peak condition M8 (Unstable Base, Compliant Foam Pad, Eyes Closed) expanded the COP path length to 321.03 cm and the prediction ellipse area to 115.42 cm^2.

By capturing an extended dynamic range, the progressive protocol is intended to scale task difficulty and elicit measurable postural adjustments, offering a potential approach to reduce ceiling effects observed in conventional Romberg-style tests.

3.2. Comparative Analysis of Sway Area Models

A key feature of the framework is the simultaneous calculation of distinct spatial models to demonstrate the methodological heterogeneity in current posturographic literature. When applied to identical datasets, the three implemented methods yielded quantitatively distinct results.

In measurement stage M6, the 95% Confidence Circle, directly sensitive to maximum radial distance, produced an area calculation of 38.90 cm^2. In contrast, the 95% Prediction Ellipse, which accounts for sway covariance and primary orientation while mitigating the influence of extreme spatial outliers, calculated a smaller area of 36.41 cm^2. Meanwhile, the Convex Hull, capturing the absolute spatial boundary of the trajectory regardless of statistical distribution, yielded an area of 45.92 cm^2. These numerical discrepancies empirically illustrate that comparing 'Sway Area' across studies without explicitly declaring the underlying mathematical model introduces substantial methodological incomparability. The OpenBalance framework addresses this by calculating and visualizing all three models simultaneously (see Figures 1 and 2), providing researchers with a transparent geometric context.

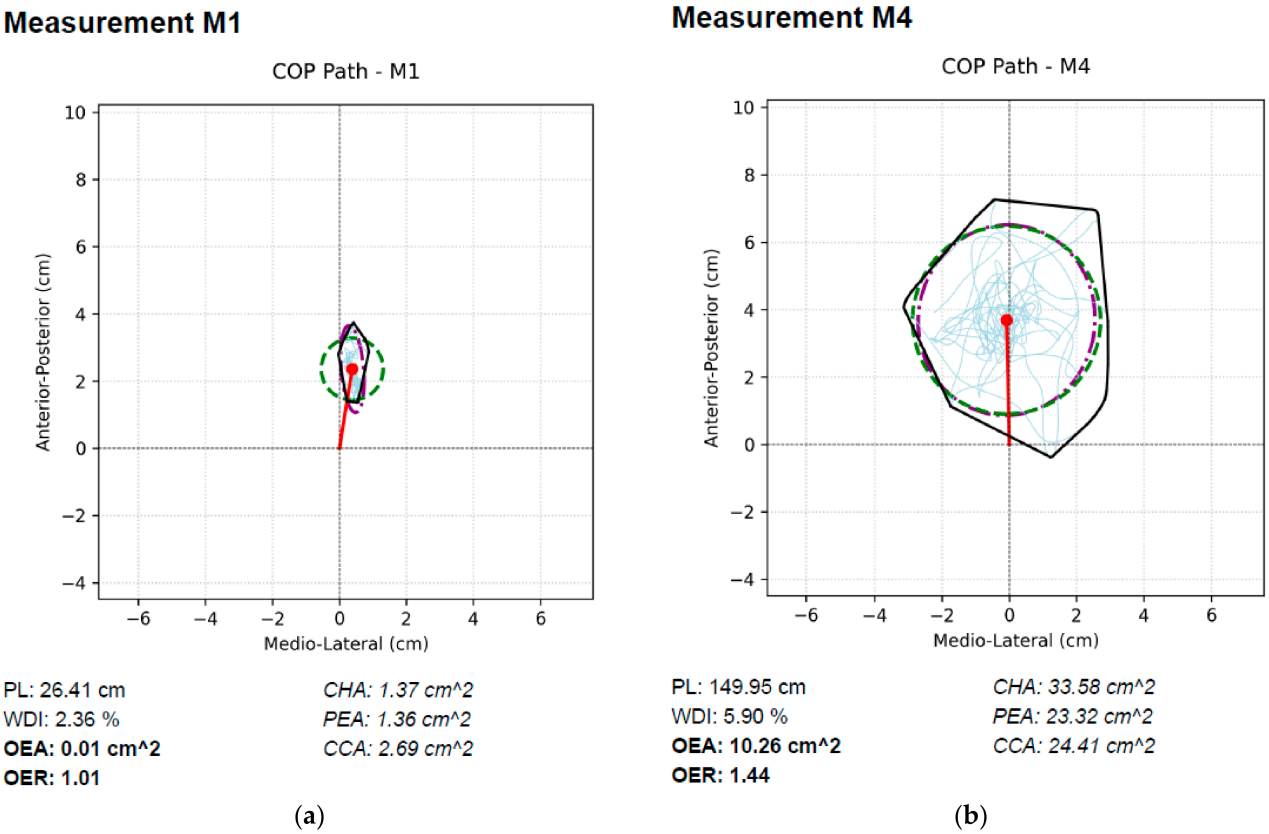


Figure 1. Visual comparison of Center of Pressure (COP) trajectories illustrating the dynamic range of the 8-stage measurement protocol. Both plots utilize identical global axis scaling to ensure direct visual comparability; stage M4 is presented alongside M1 to preserve the visual legibility of the baseline

trajectory, which would otherwise be compressed to an unresolvable point under peak M8 spatial limits. (**a**) Stage M1 (baseline condition: stable surface, eyes open) exhibits a tightly clustered COP path representing low-amplitude baseline sway. (**b**) Stage M4 (intermediate challenge: stable surface, compliant foam pad, eyes closed) displays pronounced, large-amplitude corrective movements. The spatial expansion of the Convex Hull (solid black line) relative to the 95% Prediction Ellipse (dash-dotted purple line) visually illustrates an elevated Outlier Excursion Ratio, demonstrating the progressive spatial expansion relative to static baseline conditions. Additionally, the 95% Confidence Circle is shown (dashed green line).

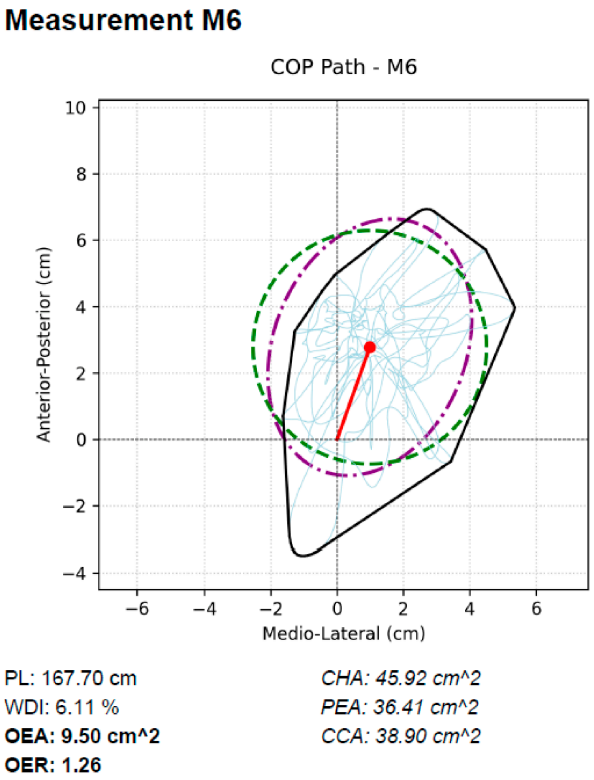


Figure 2. Comparison of three Sway Area models applied to a COP trajectory with a pronounced corrective outlier. Unlike the 95% Prediction Ellipse [purple] and Confidence Circle [green], which statistically filter the extreme excursion to capture the core sway, the Convex Hull [black] encompasses the absolute boundaries. This geometric disparity forms the basis of the Outlier Excursion Ratio.

3.3. Application and Sensitivity of the Outlier Excursion Ratio

To demonstrate the framework's ability to capture transient spatial excursions that are typically excluded by standard confidence boundaries, the Outlier Excursion Ratio was evaluated using the sample dataset. The software autonomously calculated the OER across all eight measurement stages.

Within this proof-of-concept trial, during stable measurements (e.g., M1), the OER remained at baseline level (e.g., 1.01 in Figure 1), indicating a steady, tightly clustered COP distribution where the Convex Hull closely approximated the 95% Prediction Ellipse. However, during the transition to greater instability (e.g., M4 and M6), the recorded trajectory included transient, large-amplitude excursions. Mathematically, these isolated extreme data points substantially expanded the Convex Hull without proportionally affecting the core region of the Prediction Ellipse, causing the calculated OER to increase to 1.44 and 1.26, respectively.

While large-scale clinical studies are required to establish normative data and validate the physiological relevance of this parameter, the automated extraction within the OpenBalance framework illustrates the metric's mathematical sensitivity to spatial trajectory excursions. It provides researchers with a deployable, dimensionless tool to further

investigate and quantify the spatial differences between baseline sway and transient extreme excursions.

4. Discussion

The primary objective of this study was to address the prevailing methodological inconsistencies in posturography by introducing a unified computational framework alongside a progressive assessment protocol. The contributions of this work are structured around three main pillars: software architecture, protocol design, and parameter extraction.

First, to facilitate cross-study comparability on a software level, we introduced the interoperable OpenBalance software architecture. During development, the framework was evaluated using datasets from established commercial systems (Tetrax, Tymo, AMTI, Kistler), as well as our custom OpenBalance hardware. This successful multi-vendor reader integration demonstrates the platform's software-level interoperability. Through its modular, extensible interface design, the system can be adapted via custom reader interfaces to incorporate data from various biomechanical devices that record ground reaction forces, thereby standardizing signal processing across disparate laboratory setups.

Second, to address the limited physiological challenge in standard assessments, we proposed a progressive 8-stage measurement protocol. Initial testing demonstrated the technical feasibility of this procedure, particularly the inclusion of zero-load offset measurements conducted immediately before and after the complete measurement series, which proved effective in ensuring sensor baseline stability. Furthermore, the core concept of utilizing an unstable hemispherical base (Kalotte) to elicit elevated postural demands has an empirical foundation, having been employed in previous laboratory studies at the University of Applied Sciences Technikum Wien [24]. Integrating this approach into a standardized matrix is designed to systematically scale postural challenge, offering a progressive setup intended to potentially mitigate ceiling effects in healthy cohorts.

Third, we addressed the mathematical ambiguities surrounding Sway Area calculations by introducing the Outlier Excursion Ratio. By leveraging the geometric disparity between the statistical core (95% Prediction Ellipse) and the absolute trajectory boundaries (Convex Hull), the OER mathematically quantifies extreme spatial trajectory excursions. As a dimensionless metric, it offers researchers a standardized tool to incorporate these transient spatial outliers into statistical analyses without the confounding influence of absolute baseline sway magnitude. However, while its mathematical sensitivity has been demonstrated within this framework, its specific diagnostic value and clinical relevance must be comprehensively evaluated and validated in future large-scale clinical trials.

The necessity for standardized posturographic analysis has been recognized in the recent literature. For instance, notable efforts by Quijoux et al. [25] provided a comprehensive review of COP variables alongside an open-access evaluation tool. While such platforms are valuable, they typically operate as fixed web-based upload services or rigid scripts, which limit their adaptability. Users upload data and receive a predefined output, making it difficult to modify the underlying algorithms to accommodate custom study designs, dynamic protocols, or specialized hardware formats.

In contrast, the OpenBalance architecture presented in this study shifts the paradigm from a "data-upload service" to an adaptable, modular software library. By providing a fully integrable Python package, researchers are empowered to write custom evaluation pipelines, programmatically iterate over extensive datasets, and adapt the posturographic outputs specifically to their unique measurement protocols (such as the proposed 8-stag matrix).

Furthermore, as established in our foundational hardware work [18], OpenBalance extends beyond standalone software to form an integrated processing framework. A core

objective of this framework is to facilitate standardized data ingestion and software-level interoperability across platforms. By routing data from high-cost reference commercial systems and low-cost custom platforms through an identical mathematical processing pipeline, researchers can evaluate raw signal characteristics without the confounding variables of proprietary, black-box software filtering.

Limitations

Several limitations of this proof-of-concept study must be acknowledged. First, all empirical data originate from a single healthy adult volunteer; consequently, no population-level conclusions regarding ceiling effects, protocol sensitivity, or the diagnostic value of the OER can be drawn, and the reported values serve exclusively as a technical feasibility demonstration. Second, the OER is inherently dependent on recording duration and sampling frequency, as the Convex Hull Area grows monotonically with additional samples while the Prediction Ellipse Area approaches an asymptotic limit; cross-study comparisons therefore require strictly standardized trial durations. Third, the low-pass filter cutoff interacts directly with the OER, as filtering attenuates the extreme spatial excursions the metric is designed to capture; the cutoff frequency must therefore always be reported alongside OER values. Fourth, normative OER data, test–retest reliability estimates, and validation against clinical instability outcomes are not yet available and constitute the primary objective of planned cohort studies.

5. Conclusions

This study presented OpenBalance, an open-source, modular Python framework designed to address methodological inconsistencies in posturographic signal processing and parameter extraction. By establishing standardized data input interfaces across commercial and custom force plate formats, introducing a progressive 8-stage measurement protocol, and defining the dimensionless Outlier Excursion Ratio as a spatial-magnitude metric, the framework provides an adaptable infrastructure for human balance research. Initial evaluation on a single-volunteer dataset demonstrated the technical execution and computational feasibility of the processing pipeline.

To advance this open-access framework, future work will focus on expanding the physical and graphical infrastructure. The immediate next step involves developing a dedicated Python-based Graphical User Interface (GUI) to streamline experimental data acquisition by directly recording and structuring multi-channel sensor data into the standardized OpenBalance trial schema. In parallel, the mechanical platform design is being refined for FDM (Fused Deposition Modeling) 3D printing, obviating the requirement for costly CNC machining and enabling research groups to manufacture structural components in-house. To facilitate electronic assembly, a custom printed circuit board (PCB) carrier is being developed as a pre-configured project with a commercial PCB fabricator alongside open-access CAD files, STL models, and a complete Bill of Materials (BOM).

Finally, optimized microcontroller firmware is currently being finalized to deliver a strictly constant 200 Hz data stream, minimizing timing jitter to ensure accurate frequency-domain calculations (e.g., Power Spectral Density via Fourier Transform). Ultimately, by combining a verified software library with accessible 3D-printable hardware, the OpenBalance project aims to reduce the financial and technical barriers associated with high-resolution human balance assessment.

Author Contributions: O.H.: Conceptualization, Methodology, Software, Investigation, Data Curation, Formal Analysis, Validation, Visualization, Writing—Original Draft. T.B.: Conceptualization, Methodology, Supervision, Writing—Review and Editing. A.S.: Conceptualization, Validation, Supervision, Writing—Review and Editing. M.B.: Resources, Project Administration, Supervi-

sion, Writing—Review and Editing. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki. Ethical review and approval were waived for this study because the proof-of-concept data were acquired exclusively from one of the authors as a technical system verification.

Informed Consent Statement: Informed consent was obtained from the participating author.

Data Availability Statement: The data are available in a GitHub repository, with the corresponding link becoming active upon publication. The complete computational framework and all empirical datasets (M1–M8) are openly available at the following link: https://github.com/dryride/OpenBalance (accessed on 14 October 2026).

Acknowledgments: Building upon the open-access hardware repository established in our previous work [18], this research expands the OpenBalance ecosystem into a comprehensive, scientific monorepository. To ensure maximum reproducibility and serve as a centralized hub for community-driven development, the repository structure has been converted to conform to modern software engineering standards for Python packages. During the preparation of this manuscript, the authors used Gemini Advanced (Google LLC, Mountain View, CA, USA) for the purposes of translation, language polishing, and restructuring code for the associated GitHub repository. The authors have reviewed and edited the output and take full responsibility for the content of this publication. The computational framework introduced in this paper is encapsulated within the core/openbalance directory, operating as a standalone, modular Python library. To maximize accessibility for researchers and clinicians, the package has been published on the Python Package Index (PyPI), enabling seamless installation across operating systems via the standard package manager (i.e., pip install openbalance). To distinguish between library infrastructure and execution, all high-level execution scripts—including the automated processing pipeline for the 8-stage protocol—are housed in the/examples directory. Empirical datasets are strictly categorized within the/data directory. Alongside the legacy hardware validation matrices, we have published the complete multi-channel data streams (M1 through M8) utilized to validate the OER and generate the verification plots presented in this study. Furthermore, to consolidate the scientific foundation of the project, a dedicated/publications directory has been introduced. This directory archives the peer-reviewed manuscripts, preprints, and methodological documentation for both the hardware design and the current software framework. The complete, fully integrated ecosystem remains persistently maintained and publicly accessible under the MIT License at: https://github.com/dryride/OpenBalance (accessed on 14 October 2026).

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

COP	Center of Pressure
PL	Path Length
WDI	Weight Distribution Index
ROM	Range of Motion
CHA	Convex Hull Area
PEA	Prediction Ellipse Area
CCA	Confidence Circle Area
OEA	Outlier Excursion Area
OER	Outlier Excursion Ratio
SOT	Sensory Organization Test

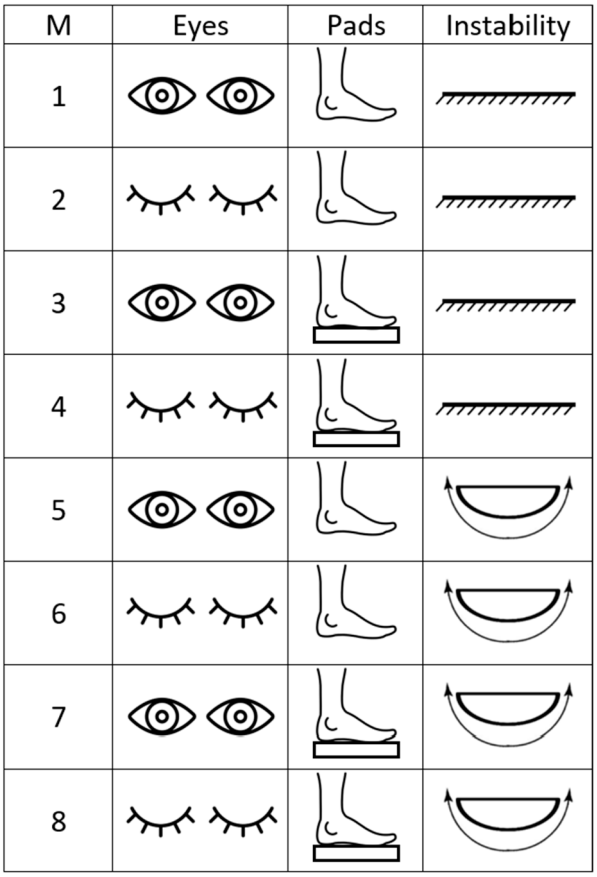
Appendix A

This appendix provides a detailed tabular breakdown of the proposed progressive 8-stage posturographic assessment protocol. The table specifies the exact environmental and physiological configurations for each of the eight sequential measurements (M1–M8). The test conditions are systematically categorized by visual input (Eyes Open vs. Eyes Closed), surface compliance (Stable vs. Compliant Foam Pads), and base stability (Flat Surface vs. Hemispherical Unstable Base). This structured, text-based overview serves as a standardized reference to ensure accurate replication of the experimental setup in clinical and biomechanical research settings.

Table A1. Conditions for the new measurement protocol.

Measurement	Eyes	Pads	Instability
1	Open	no	no
2	Closed	no	no
3	Open	yes	no
4	Closed	yes	no
5	Open	no	yes
6	Closed	no	yes
7	Open	yes	yes
8	Closed	yes	yes

Appendix B



This appendix illustrates the progressive 8-stage measurement protocol utilizing a visual pictogram matrix. The figure depicts the sequence of measurements (M1–M8) along the vertical axis, while the horizontal columns visually indicate the specific test conditions

through icons (i.e., visual state, presence of foam pads, and application of an unstable base). This graphical representation is specifically designed as a quick-reference guide for laboratory personnel and clinicians, facilitating efficient and error-free protocol execution during active data collection.

References

1. Carrick, F.R.; Hankir, A.; Zaman, R.; Wright, C.H.G. Metrological Performance of Instruments Used in Clinical Evaluation of Balance. *Psychiatr. Danub.* **2019**, *31*, 324–330. [PubMed]
2. Scoppa, F.; Capra, R.; Gallamini, M.; Shiffer, R. Clinical Stabilometry Standardization. Basic Definitions–Acquisition Interval–Sampling Frequency. *Gait Posture* **2013**, *37*, 290–292. [CrossRef] [PubMed]
3. Scoppa, F.; Gallamini, M.; Belloni, G.; Messina, G. Clinical Stabilometry Standardization: Feet Position in the Static Stabilometric Assessment of Postural Stability. *Acta Medica Mediterr.* **2017**, *33*, 707–713. [CrossRef]
4. Ruhe, A.; Fejer, R.; Walker, B. The Test-Retest Reliability of Centre of Pressure Measures in Bipedal Static Task Conditions—A Systematic Review of the Literature. *Gait Posture* **2010**, *32*, 436–445. [CrossRef] [PubMed]
5. Kohen-Raz, R. Application of Tetra-Ataxiametric Posturography in Clinical and Developmental Diagnosis. *Percept. Mot. Ski.* **1991**, *73*, 635–656. [CrossRef] [PubMed]
6. Nashner, L.M.; Black, F.O.; Wall, C. Adaptation to Altered Support and Visual Conditions during Stance: Patients with Vestibular Deficits. *J. Neurosci.* **1982**, *2*, 536–544. [CrossRef] [PubMed]
7. Maas, S.A.; Göcking, T.; Stojan, R.; Voelcker-Rehage, C.; Kutz, D.F. Synchronization of Neurophysiological and Biomechanical Data in a Real-Time Virtual Gait Analysis System (GRAIL): A Proof-of-Principle Study. *Sensors* **2024**, *24*, 3779. [CrossRef] [PubMed]
8. Curran, K.; Holland, G.; Redfern, M.S.; Cham, R. Comparing Sensory Organization Test Measures of the Bertec® Balance Advantage® CDP/IVRTM and NeuroCom® Smart EquiTest® Computerized Dynamic Posturography Systems in Young and Older Healthy Adults. *Gait Posture* **2025**, *121*, 308–314. [CrossRef] [PubMed]
9. Mancini, M.; Horak, F.B. The Relevance of Clinical Balance Assessment Tools to Differentiate Balance Deficits. *Eur. J. Phys. Rehabil. Med.* **2010**, *46*, 239–248. [PubMed]
10. Paillard, T. Plasticity of the Postural Function to Sport and/or Motor Experience. *Neurosci. Biobehav. Rev.* **2017**, *72*, 129–152. [CrossRef] [PubMed]
11. Ringhof, S.; Stein, T. Biomechanical Assessment of Dynamic Balance: Specificity of Different Balance Tests. *Hum. Mov. Sci.* **2018**, *58*, 140–147. [CrossRef] [PubMed]
12. Schubert, P.; Kirchner, M. Ellipse Area Calculations and Their Applicability in Posturography. *Gait Posture* **2014**, *39*, 518–522. [CrossRef] [PubMed]
13. Prieto, T.E.; Myklebust, J.B.; Hoffmann, R.G.; Lovett, E.G.; Myklebust, B.M. Measures of Postural Steadiness: Differences between Healthy Young and Elderly Adults. *IEEE Trans. Biomed. Eng.* **1996**, *43*, 956–966. [CrossRef] [PubMed]
14. Duarte, M.; Zatsiorsky, V.M. Effects of Body Lean and Visual Information on the Equilibrium Maintenance during Stance. *Exp. Brain Res.* **2002**, *146*, 60–69. [CrossRef] [PubMed]
15. Barber, C.B.; Dobkin, D.P.; Huhdanpaa, H. The Quickhull Algorithm for Convex Hulls. *ACM Trans. Math. Softw.* **1996**, *22*, 469–483. [CrossRef]
16. Virtanen, P.; Gommers, R.; Oliphant, T.E.; Haberland, M.; Reddy, T.; Cournapeau, D.; Burovski, E.; Peterson, P.; Weckesser, W.; Bright, J.; et al. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nat. Methods* **2020**, *17*, 261–272. Correction in *Nat. Methods* **2020**, *17*, 352. [CrossRef] [PubMed]
17. Martin, R.C. Design Principles and Design Patterns. *Object Mentor* **2000**, *1*, 597.
18. Hofstätter, O.; Bochdansky, T.; Sabo, A.; Bäckström, M. Methodological Framework for Multi-Cell Posturography Enabling Unconstrained Foot Placement and Open-Source Balance Assessment. *Sensors* **2026**, *26*, 6210. [CrossRef] [PubMed]
19. Peng, R.D. Reproducible Research in Computational Science. *Science* **2011**, *334*, 1226–1227. [CrossRef] [PubMed]
20. Winter, D.A. *Biomechanics and Motor Control of Human Movement*; John Wiley & Sons: Hoboken, NJ, USA, 2009; ISBN 978-0-470-39818-0.
21. Koltermann, J.; Gerber, M.; Beck, H.; Beck, M. Validation of Various Filters and Sampling Parameters for a COP Analysis. *Technologies* **2018**, *6*, 56. [CrossRef]
22. Shumway-Cook, A.; Horak, F.B. Assessing the Influence of Sensory Interaction on Balance: Suggestion from the Field. *Phys. Ther.* **1986**, *66*, 1548–1550. [CrossRef] [PubMed]
23. Winter, D.A. Human Balance and Posture Control during Standing and Walking. *Gait Posture* **1995**, *3*, 193–214. [CrossRef]

24. Navratil, B. Parameter der Posturographie. Master's Thesis, University of Applied Sciences Technikum Wien, Vienna, Austria, 2010.
25. Quijoux, F.; Nicolaï, A.; Chairi, I.; Bargiotas, I.; Ricard, D.; Yelnik, A.; Oudre, L.; Bertin-Hugault, F.; Vidal, P.P.; Vayatis, N.; et al. A Review of Center of Pressure (COP) Variables to Quantify Standing Balance in Elderly People: Algorithms and Open-Access Code. *Physiol. Rep.* **2021**, *9*, e15072. [CrossRef] [PubMed]

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